

DAPO Decoupled clipping & dynamic sampling for long-CoT RL

Sample a group

Math prompt + answer
q : 6 × 7 = ? a : 42

Rollout snapshot
 $\pi_{\theta_{\text{old}}}$

6 responses / prompt
Fixed for the batch

Dynamic sampling
1 0 1 Mixed → buffer All correct / all wrong → discard

Token-level clipped objective

Clip-Higher



Raise the upper clip; keep the lower clip
Clipping shapes the surrogate objective.

Verify, shape & select

Completions

... 42 <eos>	+1	✓
... 17 <eos>	-1	✗
... 42 <eos>	+1	✓

Correctness + soft length reward

Retain groups with $0 < \text{accuracy} < 1$

Mixed → buffer All correct / all wrong → discard

Reward

Group signal & trainable policy

Normalize rewards within each group
$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}$$

Share each advantage across valid tokens

Current policy
 π_{θ} Maximize the DAPO objective
No learned critic; no reference KL

Resample until the retained batch is full

$$\hat{J}(\theta) = \frac{\sum_{i,t} m_{i,t} \ell_{i,t}}{\sum_{i,t} m_{i,t}}, \quad \ell_{i,t} = \min(w_{i,t} \hat{A}_i, \text{clip}(w_{i,t}, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}) \hat{A}_i)$$

Token-level reduction

Equal weight per valid token

+ + +
- - - - - - -

m = 0 for truncated responses

Soft overlong punishment

Correctness + length penalty

